Department of Computer Science and Engineering
Bachelor of Computer Science and Engineering (BCSE)
Course (Theory/Lab) along with
Complex Engineering Problem (CEP) and Complex Engineering Activities (CEA)

- **Course Code and Title:** CSC 471 — Microprocessor Based System
- **Semester:** Summer Semester 2026
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- Which P's are addressed (Requirements: P1 is must and Some or all P2 to P7)?

Name of CEA	Addressed (Explain how it was addressed?) or —	
P1: Depth of Knowledge (Required one or more K from K3, K4, K5, K6, or K8)	**K3**	Engineering fundamentals. Discrete-time signal processing is applied throughout: anti-alias filtering precedes decimation from 200 Hz to 50 Hz, because slicing without filtering aliases the impact transient into the passband. The same Nyquist reasoning excludes UP-Fall (18.4 Hz) and UMAFall's 20.1 Hz waist channel, since neither reaches 50 Hz without fabricating signal content. Datasets are also rotated onto one gravity convention using signed permutation matrices constrained to determinant +1, which preserves gyroscope handedness.
	K4	Specialist knowledge. Full-integer INT8 quantisation, separable convolutions and global average pooling hold the network to 7,947 parameters and 22.5 KB. The work reports a non-obvious specialist result: instance normalisation placed inside the computation graph costs 30.95 points of macro-F1 under INT8, while the identical operation in the preprocessing pipeline costs nothing, because per-window mean and variance produce tensors whose dynamic range a single per-tensor quantisation scale cannot represent.
	K5	Engineering tools. TensorFlow and Keras for training, TensorFlow Lite Micro for embedded inference, Kaggle T4 GPU sessions for the experiments, and the Arduino toolchain for the ESP32. Preprocessing is defined once in a version-controlled shared library imported by both the training scripts and the firmware generator, so the two cannot drift apart.
	K6	Engineering practice. The system was built and validated on real hardware: an ESP32 DevKit V1 with an MPU6050 over I2C at 400 kHz, configured to

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		±16 g and ±2000 dps, sampled by a hardware-timer interrupt at 50 Hz, driving a buzzer and status LED, with on-device instrumentation for inference latency and tensor-arena usage.
	K8	Research literature. Four public datasets were audited against their source publications, and the results are positioned against the KFall benchmark (Yu et al., 2021), TinyFallNet (2023), PreFallKD (2023) and the cross-dataset study of Silva et al. (2024). Because that last work already establishes that a cross-dataset gap exists, the contribution here is deliberately narrowed to which mitigations are affordable on a microcontroller.
P2: Conflicting Requirements	Addressed. This is the report's central engineering tension. Lead time and false-alarm rate conflict directly: requiring k consecutive windows before alarming cuts trial-level false positives from 538 to 26 out of 2,729 ADL trials, but each extra window costs one 0.5 s stride of lead time, and the mean lead of 663 ms is barely longer than a single stride. Model size, latency and accuracy conflict likewise: the proposed network gives up about two points of macro-F1 against the best comparator for a thirteen-fold reduction in parameters. Both trade-offs are quantified rather than asserted.	
P3: Depth of Analysis	Addressed. No textbook procedure answers the questions posed. The work required a tiered cross-validation design (5-fold subject-grouped for comparisons, full leave-one-subject-out for the headline result, and leave-one-dataset-out for the generalisation claim), an operating-point sweep over decision threshold and agreement count, and a null-controlled experiment to establish that the SisFall Enhanced annotations were unrecoverable — recovery peaked at 22.5 % against a time-reversed control and fell under harder filtering, which is what makes 22.5 % a ceiling rather than a waypoint.	
P4: Familiarity of Issues	Addressed. Cross-dataset generalisation in wearable fall detection is an open research question rather than a routine exercise, and several problems encountered are not documented in the literature: that a dataset's gravity axis convention can masquerade as domain shift, and that the placement of instance normalisation relative to the quantisation boundary decides whether an INT8 model functions at all.	
P5: Extent of Applicable Codes	Partially addressed. No single standard governs a research prototype of this kind, but the work follows established practice: IEEE conference format for reporting, I2C bus specifications for the sensor interface, TensorFlow Lite Micro conventions for embedded inference, and the reproducibility norms of the field — fixed random seeds, published preprocessing constants and fold-level result files released publicly.	
P6: Extent of Stakeholders	Addressed. Older adults at risk of falling are the end users, with caregivers and clinicians as secondary stakeholders whose tolerance for false alarms determines whether a device is worn at all. The BDT 1,450 bill of materials responds directly to affordability constraints in the Bangladeshi context, and the false-alarms-per-hour analysis exists precisely because a device that alarms hundreds of times an hour will be removed and never used, however accurate it is.	
P7: Interdependence	Addressed. The subproblems cannot be solved independently. The sampling rate fixes the input tensor, which fixes model size and inference latency, which fixes the feasible	

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	stride, which bounds the achievable lead time and the false-alarm rate. The choice of normalisation strategy simultaneously determines cross-dataset transfer — it is the only adaptation method that helped — and whether the model survives quantisation. Changing any one of these forces the others to be revisited.

- Which A's are addressed (Requirements: Some or all A1 to A5)? [Only Relates to PO(j)]

Attributes of CEA	**Addressed** (Explain how it was addressed?) **or —**
A1: Range of Resources	Addressed. Four public datasets totalling over 250,000 labelled windows; cloud GPU compute on Kaggle T4 accelerators; embedded hardware comprising ESP32, MPU6050, TP4056 charger and LiPo cell; software toolchains spanning TensorFlow, TensorFlow Lite Micro and Arduino; and the published research literature of the field.
A2: Level of Interaction	Addressed. The work required reconciling requirements across machine learning, embedded systems and signal processing, and coordination within a five-member group across data preparation, model training, firmware development, hardware assembly and report writing, using a shared version-controlled repository as the coordination mechanism.
A3: Innovation	Addressed. Two findings appear not to have been reported previously: that instance normalisation must sit on the float side of the quantisation boundary, worth 30.95 points of macro-F1; and that the domain-adaptation ladder inverts, with the cheapest method the only one that improves transfer while CORAL and adversarial adaptation both degrade it. The argument that window-level specificity is close to meaningless for a continuously running detector is also not made elsewhere in this literature.
A4: Consequences to Society and Environment	Addressed. Falls are a leading cause of injury-related death among older adults, and a pre-impact alarm enables protective action rather than only post-hoc alerting. At BDT 1,450 the device is affordable in a low-resource setting. Running a 22.5 KB model on a microcontroller also avoids the energy cost and the privacy exposure of streaming continuous motion data to a server, since all inference happens on the device.
A5: Familiarity	Addressed. The work goes well beyond routine coursework: auditing public datasets against their source publications and finding four material defects, designing leakage-free evaluation protocols, diagnosing a silent quantisation failure before deployment, and running a neural network on a microcontroller with measured resource usage.

Comments: ____

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